# Qijin Xu

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## Summary

Lead AI/ML Engineer and Software Developer with 5+ years of experience; architected and productionized AI platforms and agents that: automated 90% of workloads and reduced manual labour by 50% at 50,000+ user scale, delivered 525% ROI; in ML, achieved a 321% lift in fraud-loss prevention vs a top-five bank benchmark.

## Technical Skills

**Languages:** Python, Java, Go, SQL(TSQL, PostgreSQL), JavaScript, C++
**AI/ML:** PyTorch, LangGraph, Retrieval-Augmented Generation (RAG), agent workflows / agents, Model Context Protocol (MCP), Large Language Models (LLMs), FAISS, Pinecone, OpenAI API, vLLM, JAX
**Data/Streaming:** Apache Spark, Kafka, Airflow, Data Lineage (OpenLineage), Redis
**Cloud/Infra and Systems:** Azure, GCP, AWS, Docker, Kubernetes, Terraform, Microservices, gRPC, CI/CD

## Experience

### CGI Inc. — Toronto, ON

**Lead AI/ML Engineer (Tech Lead), AI Center of Excellence** — Jan 2025 - Present

- Lead an AI agent for a proprietary coding language, with a LangGraph dual-state dynamic workflow, vLLM-served LLMs with RAG and EBNF structured outputs, PII filtering, documentation, and business-rule extraction; projected 50% labor reduction, maintain legacy scripts, and help future migration readiness.

**Data Architect and Engineering Lead, Senior Consultant** — Aug 2023 - Dec 2024

- CGI Impact Award 2024: Architected and shipped an LLM-powered enterprise RAG chatbot (LangChain, Azure/AWS/GCP); achieved approx. 90% autonomous resolution for a 50K+ user base and delivered 525% ROI in 12 months. PRODUCT PAGE, COVERAGE
- Built an OpenLineage-based lineage tool auto-documenting extensible multi-source flows, with LLM-assisted root-cause analysis suggestions, reduced investigation time across 3 incidents per month.
- Automated AxiomSL rule updates with LLM at RBC; about 80% manual cut.

**Data Engineer, Consultant** — Aug 2021 - Jul 2023

- Managed Canada's largest automobile-insurance policy-history database (2TB big on SQL server), holding 99.9% uptime and less than 200 ms median queries.
- Built production ETL/ELT pipelines for 22+ micro data marts (less than 6 tables each; monthly SLAs) powering finance, analytics and delivery department; cut report runtime from 30 min to 5 min.
- Shipped a machine learning (Gradient-Boost + Logistic Regression) loss-ratio model that lifted recovered fraud losses 321% versus a top-5 bank benchmark, batch-scoring in a Python/T-SQL pipeline.

### Elder Laboratory, York University — Toronto, ON

**Research Assistant, Computer Vision** (part-time; concurrent) — May 2020 - Aug 2021

- Developed AttentiveVision, a computer-vision pipeline for feature extraction on hockey broadcast video in Python (PyTorch, TensorFlow, OpenCV), training a CNN-LSTM stack that reached mAP greater than or equal to 70% on event detection across 18 hours of game video.
- Engineered a real-time player tracking system using a Kalman-filter tracker and OpenPose, delivering less than 100 ms latency at 30 FPS, capturing pose and momentum for live speed & heat-map analytics.
- Created an event-detection and annotation (Goal, Timeout, etc.) model for sport game videos, cutting manual labelling effort and raising dataset reliability to Cohen's kappa greater than or equal to 0.92.

## Education

**York University** — Toronto, ON
**Bachelor of Science in Computer Science, Specialized Honours** — Sep 2017 - Apr 2020

GPA: 3.9/4.0; First in Class; Dean's List (all semesters)

## Certifications

Microsoft Certified: Azure Solutions Architect Expert, Google Cloud Professional Cloud Developer